

ALESSIO MARTINI

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Theoretical physics MSc student specializing in stochastic processes, non-equilibrium statistical mechanics and complex systems. Seeking a quantitative research role centered on medium- to long-horizon strategy and risk modelling, rather than high-frequency trading.

EDUCATION

Master of Science in Theoretical Physics | University of Amsterdam

Amsterdam, Netherlands

GPA: 8.76/10

Sep 2024 – Jul 2026

- **Thesis:** *Continuous Non-Invertible Symmetries in the 3d Complex Scalar Boson* — Advisor: Diego Hofman.
Constructive extension of non-invertible symmetry defects via the Symmetry TFT framework.
- **Relevant Courses:** Non-equilibrium Statistical Physics (Langevin & Fokker-Planck equations, stochastic processes), Machine Learning for Physics, Hydrodynamics, Quantum Information Theory & Computing.

Bachelor of Science in Physics | University of Milano Bicocca

Milano, Italy

Oct 2020 – Nov 2023

- **Thesis:** *Anomalies in Quantum Mechanics* — Advisor: Noppadol Mekareeya.
Analysis of scale symmetry's breakdown in a 2d Quantum Mechanical model.
- **Relevant Courses:** Computational Physics (Monte Carlo methods, numerical integration in C), Statistical Data Analysis (parameter estimation, hypothesis testing, error propagation), Complex and Functional Analysis.

TECHNICAL & METHODOLOGICAL COMPETENCIES

- **Mathematical Foundations:** Probability Theory, Advanced Linear Algebra, Variational Calculus, Differential Geometry.
- **Stochastic & Statistical Modelling:** Stochastic Differential Equations, Non-Equilibrium Statistical Mechanics, Hydrodynamic Modelling, Classical & Quantum Game Theory, Information Theory.
- **Computational Methods:** Monte Carlo Methods, High-Dimensional Numerical Integration, Finite-Difference ODE Solvers (Runge-Kutta, Leap-frog), Root-Finding (Newton, Bisection).
- **Programming & Tools:** Python (NumPy, SciPy, Pandas, StatsModels, **PyTorch in-progress**), Wolfram Mathematica, C, C++, SQL.

QUANTITATIVE & COMPUTATIONAL PROJECTS

Stochastic Processes for Financial Time Series (Python, coursework-based)

- Implemented **fractional Brownian motion** via Cholesky factorization of a 1000×1000 covariance kernel across five Hurst exponents ($H = 0.1-0.9$), validating the variogram scaling $V(\tau) \propto \tau^{2H}$ (sub- vs. super-diffusive) over 100 realizations each.
- Implemented the **Ornstein-Uhlenbeck** process (10^5 steps, 100 realizations) and validated its mean-reverting autocorrelation and plateauing variogram against closed-form theory, in contrast to fBM's unbounded growth.
- Verified monofractal scaling of fBM ($H = 0.7$, 32,768 points) via moment ratios $M_q(\tau)/\sigma(\tau)^q$ matching theoretical Gaussian constants to within 0.1–0.2% for $q = 1-4$, then measured empirical scaling exponents on a Multifractal Random Walk — the same diagnostic used to detect fat-tailed, intermittent market returns.

Fat-Tailed Statistics & Stochastic Dynamics (Python, MSc coursework)

- Simulated overdamped **Langevin** dynamics for a free particle and recovered the diffusion constant's linear temperature dependence (fitted slope 1.008 vs. theoretical $D = 1$, $< 1\%$ deviation across $T = 1-100$); extended to a quartic double-well potential and characterized barrier-crossing statistics via first-passage-time distributions across varying barrier height Δ and T , exhibiting the exponential Δ/T dependence predicted by **Kramers' escape law**.
- Sampled sums of **Pareto-tailed** random variables (tail index $\mu = 3$, 10,000 samples per N) across $N = 1-1000$ and recovered the theoretically predicted rescaling exponent $\alpha = 1/2$ via a systematic scan ($\alpha \in [0.40, 0.60]$) for best data collapse — the diagnostic used to identify the crossover from fat-tailed to Gaussian (**CLT**) behavior in return distributions.

PROFESSIONAL EXPERIENCE

Teaching Assistant & Peer Mentor | University of Amsterdam

Amsterdam, NL

MSc Quantum Field Theory I & II (Diego Hofman)

Sep 2025 – Jul 2026

- Corrected two homework sets and the final exam across both courses, for a cohort of ~ 90 students.
- Authored original session notes: worked exercises, theory summaries, and supplementary material proposed independently.

Additional Experience

Italy

Private Tutor, Advanced Mathematics & Physics (Aug 2022 – Jun 2024) · Lead Waiter, Caffè Mattà, Cremella (Jun 2020 – Feb 2022)

· Gymnastics & Calisthenics Instructor, Milano Gym Lab & ASD Sovico (Sep 2018 – Jun 2020)

MISCELLANEOUS

- **Puzzles & Competitions:** Jane Street Monthly Puzzles solver (August 2026).
- **Languages:** Italian (Mother Tongue), English (C1 Proficient User - IELTS Certification).
- **Volunteering & Interests:** Bartender & waiter, Downtown Hostel Bar (Copenhagen, 2022); Piano, Guitar, Clarinet; Artistic gymnastics, Krav Maga; Interested in every form of Art.